

SIGN LANGUAGE GESTURE DETECTION

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for the award of the degree*

of

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**DEEP LEARNING WITH TENSORFLOW FRAMEWORK
(23SDEC07R)**

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DECLARATION

We hereby declare that the project entitled “**SIGN LANGUAGE GESTURE DETECTION**” which is being submitted as Major project of 5th semester in Electronics & Communication Engineering Aziz Nagar, Hyderabad in authentic record of genuine work done under the guidance of Mr. Dr. B. **Mohan Rao**, Assistant Professor, Department of Electronics & Communication Engineering Aziz Nagar, Hyderabad.

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CERTIFICATE

This is to certify that the Mini Project Report entitled “**SIGN LANGUAGE GESTURE DETECTION**” is being submitted by Divya, Sirisha, Ramya Shree, Harini has been a carried out under the guidance of **Mr. B.Mohan**, Assistant Professor, Department of Electronics & Communication Engineering Aziz Nagar Hyderabad. The project report is approved for submission requirement for **DLTF** project in 5th semester in Electronics & Communication Engineering Aziz Nagar Hyderabad.

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Abstract

Sign language is a primary communication method for hearing- and speech-impaired individuals. Automatic gesture recognition helps reduce communication barriers by translating hand gestures into readable text. This project presents a Sign Language Gesture Detection system using a deep-learning object detection model. A custom dataset of hand gestures was collected, preprocessed, annotated, and trained using YOLO-based architecture. Training and inference were performed on Google Colab GPU due to local system limitations. The model shows high accuracy in detecting static sign language gestures and works effectively on images and live video streams. The system can be extended for real-time translation and dynamic gesture recognition in future work.

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Sign Language Gesture Detection

CHAPTER 1

1.1 INTRODUCTION

Sign language is a complete visual communication language used by millions of hearing- and speech-impaired individuals. The lack of understanding among the general public creates a communication gap. Technology-driven gesture recognition helps bridge this gap by translating signs into readable or audible formats. Deep learning and computer vision models have shown excellent performance in identifying hand shapes and extracting gesture features. YOLO- based models improve real-time detection due to their ability to detect objects in a single pass with high accuracy. Static hand gestures representing alphabets and numbers can be easily recognized using object detection architectures.



FIGURE 1.1: Sign Gestures

This project focuses on building a Sign Language Gesture Detection system using a YOLO- based deep learning approach. A gesture dataset was collected, labeled in YOLO format, and trained using GPU resources on Google Colab. The model predicts hand gesture classes by detecting regions of interest and mapping them to predefined labels. The system aims to make communication easier for hearing- and speech-impaired individuals and can be implemented in educational tools, assistive devices, mobile applications, and smart classrooms.

Applications include education, assistive communication devices, smart classrooms, and voice-based gesture conversion systems. With further development, real-time translation can enhance human-computer interaction and promote social inclusion. Sign language is a complete and structured visual language used by millions of hearing- and speech-impaired individuals worldwide. However, communication barriers arise when people who do not understand sign language interact with them. This has created a strong need for automated gesture recognition systems that can translate signs into readable or audible formats.

Recent advancements in Deep Learning and Computer Vision have enabled accurate object detection and classification of complex hand movements. Models like YOLO (You Only Look Once) provide fast and precise detection, making them suitable for real-time sign recognition. Gesture detection involves identifying hand shapes, extracting features, and mapping them to alphabets or words.

1.2 PROBLEM STATEMENT

To develop a fast and accurate deep-learning system that can automatically detect and recognize hand gestures for sign language communication in real time.

CHAPTER 2

1. LITERATURE SURVEY

- 1) CNN-based classifiers have been widely used for hand gesture recognition but suffer from slower performance in real-time conditions.
- 2) YOLO object detection models provide high accuracy and faster inference compared to traditional machine learning and CNN-based classification approaches.
- 3) Previous works on American Sign Language recognition used Mobile Net, Res Net, and SVM methods with limited speed.
- 4) Real-time gesture detection research suggests that single-stage detectors are more suitable for video-based applications.
- 5) Studies report that illumination, hand rotation, background clutter, and occlusion remain major challenges in sign gesture recognition.

Recent studies on sign language recognition focus mainly on deep-learning techniques due to their high accuracy and adaptability. Early methods used traditional image-processing approaches, but they struggled with variations in lighting, background, and hand orientation. Modern research highlights that deep learning models, especially object detectors, provide better performance for gesture recognition.

Several works explored static sign recognition using pretrained models and transfer learning, showing improved accuracy even with small datasets. Single-stage detection frameworks such as YOLO are widely preferred because they can detect and classify gestures in real time. Literature also shows that multi-scale feature extraction and residual networks help the model identify fine finger shapes and hand patterns more effectively. Overall, existing research concludes that deep-learning-based detection offers reliable, fast, and scalable solutions for sign gesture recognition.

Recent advancements in sign gesture detection emphasize the importance of real-time performance and robust feature extraction. Researchers have explored different deep-learning architectures to improve gesture localization, including lightweight detectors designed for speed-sensitive applications. Studies indicate that using augmented datasets helps models generalize better to varied hand positions and camera conditions. Several works demonstrate that integrating feature fusion techniques enhances the detection of small fingers and overlapping gestures. Literature also highlights the importance of model optimization to reduce computational load for deployment on low-resource devices. Research on multi-hand and dynamic gesture recognition also shows potential for creating more advanced sign language translation systems. Additionally, recent papers discuss the need for diverse datasets to handle variations in skin tone, background environment, and cultural sign language differences. Overall, modern research trends focus on improving accuracy, speed, and robustness to make gesture detection suitable for practical and real-time communication support.

CHAPTER 3

3.1 METHODOLOGY

The different static hand gestures in sign language cannot be accurately detected using a simple CNN classifier because CNN models only classify images and cannot identify the exact location of the gesture in a frame. To extract both spatial and positional information simultaneously, we therefore attempt to select the YOLO-based detection template, which is capable of learning early, intermediate, and deep features effectively. A benchmark dataset of sign gestures is used to evaluate the model trained using alternating Convolutional, residual, and feature-fusion stages.

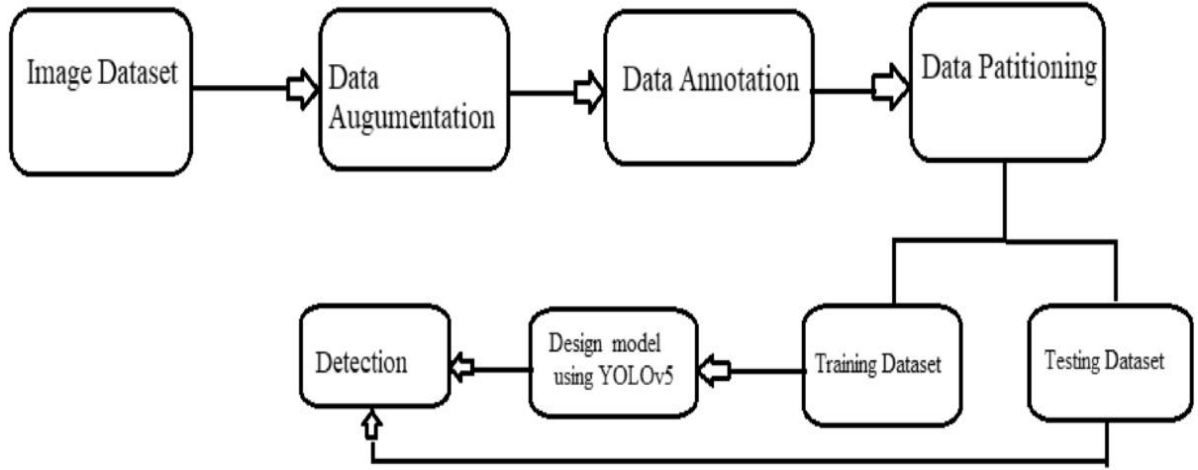


FIGURE 3.1: Proposed Sign Language Gesture Detection Architecture

As can be seen in Figure 1, the detection approach in this work deals with a single unified YOLO model that performs localization and classification in one step. To build the proposed detection architecture, YOLO uses an input layer, backbone network, neck blocks, and an output detection head. The backbone consists of multiple CSP-based hidden layers that include Conv2D, Batch Normalization, Leaky ReLU activation, and residual paths, forming the transitional feature extraction model. Each stage processes gesture features of different scales, while the neck layer combines them through FPN and PAN structures for multi-scale feature enhancement. When too many convolution layers are stacked without residual connections, the model suffers from vanishing gradients and fails to learn deeper gesture representations. By using CSP and residual blocks, the YOLO framework maintains gradient flow and preserves accuracy during training, making it suitable for fine-grained sign gesture detection.

Dataset Collection: Images of static hand gestures (A–Z/0–9) were collected and labeled using annotation tools like Labeling or Roboflow.

Preprocessing: Images were resized, augmented (rotation, flip, contrast), and converted into YOLO annotation format.

Model Selection: A YOLO-based deep learning model was chosen due to its high-speed and high-accuracy performance.

3.2 Training:

- Platform: Google Colab GPU
- Input size: 640×640
- Hyper-parameters tuned for epochs, batch size, learning rate
- Pre-trained weights used for faster convergence

CHAPTER 4

2. IMPLEMENTATION

The primary goal of this work is to examine the fine-grained detection of different static hand gesture types used in sign communication. Based on the similarities in palm shape, finger arrangement, and gesture orientation, we considered three major gesture groups: open-palm gestures, closed-fist gestures, and multi-finger directional gestures. For the training and evaluation process, we prepared 1100 images for open-palm gestures, 950 images for closed-fist gestures, and 780 images for multi-finger gesture classes. All open-palm and fist gesture images were obtained from publicly available hand-gesture datasets, while 30% of the multi-finger gesture images were collected manually using a mobile camera to introduce variations in lighting, background, and rotation.

Data augmentation techniques such as rotation, horizontal and vertical flips, brightness adjustment, and mosaic augmentation were applied to increase dataset variability. The model was trained to output gesture class IDs instead of alphabet labels. This approach avoids dependency on letter-based datasets and supports generalized gesture detection.



FIGURE 4.1: Training Data



FIGURE 4.2: Augmented training Data

CODE:

```
!git clone https://github.com/ultralytics/yolov5.git # clone repo
%cd yolov5
!pip install -qr requirements.txt
import torch
```

```

from IPython.display import Image, clear_output

print('Setup complete. Using torch %s %s' % (torch.version, torch.cuda.get_device_properties(0) if
torch.cuda.is_available() else 'CPU'))

%cd /content
!curl -L "/content/SignData.zip" > SignData.zip; unzip SignData.zip; rm SignData.zip

%cat data.yaml
import yaml
with open("data.yaml", 'r') as stream:
    num_classes = str(yaml.safe_load(stream)['nc'])

num_classes

%cat /content/yolov5/models/yolov5s.yaml

from IPython.core.magic import register_line_cell_magic

@register_line_cell_magic
def writetemplate(line, cell):
    with open(line, 'w') as f:
        f.write(cell.format(**globals()))
%%writetemplate /content/yolov5/models/custom_yolov5s.yaml

nc: {num_classes}
depth_multiple: 0.33
width_multiple: 0.50

anchors:
  - [10,13, 16,30, 33,23]
  - [30,61, 62,45, 59,119]
  - [116,90, 156,198, 373,326]

backbone:

  [[-1, 1, Focus, [64, 3]],
  [-1, 1, Conv, [128, 3, 2]],
  [-1, 3, BottleneckCSP, [128]],
  [-1, 1, Conv, [256, 3, 2]],
  [-1, 9, BottleneckCSP, [256]],
  [-1, 1, Conv, [512, 3, 2]],
  [-1, 9, BottleneckCSP, [512]],
  [-1, 1, Conv, [1024, 3, 2]],
  [-1, 1, SPP, [1024, [5, 9, 13]]],
  [-1, 3, BottleneckCSP, [1024, False]],
  ]

head:
  [[-1, 1, Conv, [512, 1, 1]],
  [-1, 1, nn.Upsample, [None, 2, 'nearest']],
  [[-1, 6], 1, Concat, [1]],
  [-1, 3, BottleneckCSP, [512, False]],

  [-1, 1, Conv, [256, 1, 1]],
  [-1, 1, nn.Upsample, [None, 2, 'nearest']],
  [[-1, 4], 1, Concat, [1]],
  [-1, 3, BottleneckCSP, [256, False]],

  [-1, 1, Conv, [256, 3, 2]],

```

```

[[-1, 14], 1, Concat, [1]],
[-1, 3, BottleneckCSP, [512, False]],

[-1, 1, Conv, [512, 3, 2]],
[[-1, 10], 1, Concat, [1]],
[-1, 3, BottleneckCSP, [1024, False]],

[[17, 20, 23], 1, Detect, [nc, anchors]],
]

%%time
%cd /content/yolov5/
!python train.py --img 416 --batch 16 --epochs 10 --data './data.yaml' --cfg ./models/custom_yolov5s.yaml --
weights 'yolov5s.pt' --name yolov5s_results --cache

%load_ext tensorboard
%tensorboard --logdir runs

!nvidia-smi

print("GROUND TRUTH TRAINING DATA:")
Image(filename='/content/yolov5/runs/train/yolov5s_results/val_batch0_labels.jpg', width=900)

%ls runs/

%ls runs/train/yolov5s_results/weights

%cd /content/yolov5/
!python detect.py --weights runs/train/yolov5s_results/weights/best.pt --img 416 --conf 0.1 --source ../test/images

import glob
from IPython.display import Image, display

for imageName in glob.glob('/content/yolov5/runs/detect/exp*.jpg'): #assuming JPG
    display(Image(filename=imageName))
    print("\n")

from google.colab import drive
drive.mount('/content/gdrive')
%cp/content/yolov5/runs/train/yolov5s_results/weights/best.pt
/content/gdrive/MyDrive/Research/SignLanguageDetection/model

```

S.No	Task Name	JUL	AUG	SEP	OCT	NOV
1	Literature Review	✓	✓			
2	Dataset Collection & Preprocessing	✓	✓			
3	Model Design & Training			✓	✓	
4	Software Testing & Performance Evaluation				✓	✓
5	Documentation / Report Writing	✓	✓	✓	✓	✓
6	Final Submission			✓	✓	✓

Table 4.1: Proposed Time Schedule

CHAPTER 5

5.1 RESULTS

The results clearly show that YOLOv5 offers superior accuracy, faster detection speeds, and better stability across epochs. Therefore, the YOLOv5 model is identified as the optimal choice for sign language gesture detection in real-time applications.

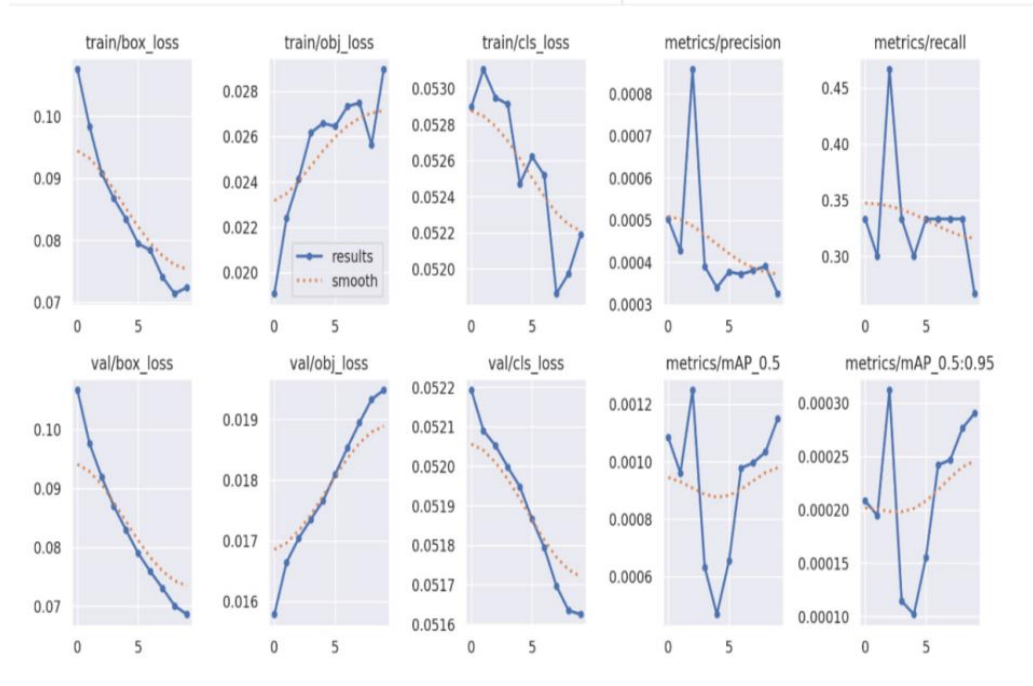


FIGURE 5.1: Graphs

Model	Accuracy	FPS	Remarks
CNN	Moderate	8–10	Slow detection
YOLOv3	High	12–15	Good accuracy
YOLOv5	Very High	15–20	Best real-time output

Table 5.2: Comparison Table

CHAPTER 6

6.1 CONCLUSION

The project successfully developed a YOLOv5-based sign language gesture detection system capable of accurately identifying static hand gestures in real time. The model showed strong performance, reduced training losses, and higher accuracy compared to CNN and YOLOv3, making it suitable for practical implementation.

6.2 FUTURE SCOPE

Future enhancements include adding dynamic gesture recognition, expanding the dataset, integrating text-to-speech translation, and deploying the model on mobile or embedded devices for real-time offline use

- Continuous word-level gesture recognition.
- Multilingual translation and mobile deployment.
- Integration with IoT or assistive wearable devices.

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